

Register Selection and the Profile-Contingency Boundary

Geist Atlas · Module 13 · Ontology, State, and the Limits of Interior Modeling

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[**SCOPE-BOUND**] Supported but scope-bound · **Family:** Ontology, State, and the Limits of Interior Modeling

Claim: A pre-registered two-family test does not confirm a profile-contingent benefit from adaptive register selection: Terra fails the binding profile-manipulation gate, Sonnet passes the gate but returns a null interaction, and the field selector is closed on this evidence.

A 120-dialogue fixed-horizon confirmatory study and provenance stress test. It separates endpoint level from profile separability, retains a bounded hostile-register safety hypothesis, and limits the result to one simulated detective world and whole-stack model families rather than an isolated tutor-model or human-learning claim.

Derived view of `paper-full-2.0.md` v3.0.227 — §6.17. The paper is canonical; edit it there, not here.

6.17 Register Selection Does Not Produce a Confirmed Profile-Contingent Benefit Across Two Model Families (Pre-Registered Gate)

The register-policy line asked whether a tutor should change its speaking stance according to a learner profile, rather than whether one register is best in aggregate. A frozen confirmatory

*This sentence is the only one actually authored by Liam Magee in this paper.

design crossed three policies (`bland`, the adaptive `field` selector, and a fixed `negative` stress register) with four v3 simulated-learner contracts (`diligent`, `affective_resistant`, `false_memory`, `proof_skipper`) at $n = 5$ per cell on `world_005_marrick`, once on an all-Terra stack and once on an all-Sonnet-5 stack: 60 dialogues per family, 120 total. The primary endpoint was mechanically computed proof-DAG coverage at learner turn 16. Each family also had to pass its own in-run profile-discrimination gate — average pairwise cosine < 0.85 and maximum similarity to `diligent` < 0.90 — before any policy-by-profile interaction could confirm. A block that failed this manipulation gate was instrument-invalid for the family claim, regardless of an attractive cell contrast.

Whole stack	Selected traces	Average profile cosine	Maximum similarity to diligent	Frozen manipulation gate	Interaction verdict
Terra (<code>gpt-5.6-terra</code> at all four seams)	60	0.812	0.912	fail	instrument-invalid
Sonnet 5 (Sonnet at all four seams)	60	0.645	0.694	pass	null

Strict result: no family confirmation; no two-family claim. Terra’s only supported policy-by-profile contrast was produced by the fixed hostile floor: the `negative` contrast for `affective_resistant` relative to its diligent contrast was -0.334 (95% bootstrap interval $[-0.468, -0.167]$). It ran opposite to the pre-registered beneficial-crossing direction and the family failed the binding manipulation gate because its maximum similarity to diligent was 0.912. It is therefore descriptive evidence of a possible stack-bounded contraindication, not a register-selection positive. Sonnet passed the frozen cosine gate, but every primary interaction interval crossed zero. The adaptive `field` policy confirmed no interaction on either stack. Excluding the four Sonnet rows whose runs ended after their valid turn-16 assessment but before full secondary closure leaves 56 rows and the same null verdict.

The comparison is deliberately read as a whole-stack boundary, not a tutor-model isolation. Tutor, automated learner, classifier, and learner-record extractor changed together between Terra and Sonnet; the coverage aggregate is deterministic, but the record from which it is computed still passes through an LLM extraction seam. The large cross-stack level difference can therefore arise from tutor conduct, learner conduct, extraction calibration, or their interaction. What closes is the general claim that this adaptive selector produces a profile-contingent fixed-horizon benefit on this apparatus. The register palette remains a useful implementation vocabulary, and Terra’s hostile-register collapse remains a safety hypothesis, but neither licenses another selector-rescue block.

Status and caveats (per §5.12.6). Pre-registered and complete at $n = 5$ per cell; one detective world, simulated contract learners, one operational fixed-horizon endpoint,

and no human-learning claim. Terra has complete secondary traces for all 60 selected rows; Sonnet has 60 primary-complete rows, 56 with full secondary closure, and four primary-only rows. One instrument boundary discovered after the fact: the Sonnet family block ran before the safe-mode-v1 claude-CLI context-isolation fix, so its speaking-tutor calls ambiently loaded ~16k tokens of repository context (the codex/Terra path was always isolated). This is a whole-family transport condition, uniform within the block — it cannot manufacture the within-family register contrasts, but it adds one more uncontrolled difference to the already whole-stack Terra-vs-Sonnet comparison above, and Sonnet rows from this block must not be pooled with post-fix runs (check run-header provenance before any cross-run pooling; the §6.18-addendum Step 4 run is entirely post-fix). Reproduction is zero-call and hash-verified: the explicit final-row selection, 5,000-draw bootstrap (seed 20260713), profile gates, QA matrix, archive lineage, and strict verdict live in `scripts/analyze-register-confirmatory-step2.js`, `exports/register-confirmatory-evidence/final/`, and `config/adaptive-tutor-evidence/tutor-stub-reg`. The next experiment changes both delivery channel and intervention grain: point-of-action coaching and compiled constraints are a successor test, not a retry of the selector.

Neighbouring modules: Module 7 — Ontology, State, and the Failure of Rich Learner Interiors; Module 3 — The Adaptive-Responsiveness Null; Module 12 — Composition: Instructions Converge, Signal Accumulates; Module 4 — Provable Discourse and Process Tracing for Agent Architectures.